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Completing completions: the Karoubi envelope of the copower completion

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Free completions under limits and colimits are key constructions in category theory. In this work, motivated by computer science applications, we describe a connection between three distinct completions: the free coproduct completion (Fam-construction), the free copower completion, and the split idempotent completion (Karoubi envelope).

The main result we prove is that, for any category $\mathbb{C}$, the constant family inclusion extends to an equivalence

$$\mathrm{Kar}\big(\mathrm{Copow}(\mathbb{C})\big) \simeq \mathrm{Fam}(\mathbb{C})$$

if, and only if, $\mathbb{C}$ is Karoubi complete and every family of objects admits a *common retract host*: for a family $\{C_i\}_{i\in I}$ of objects in a category $\mathbb{C}$, morphisms

$$C_i \xrightarrow{s_i} C \xrightarrow{r_i} C_i,$$

such that $r_i \circ s_i = \mathrm{id}_{C_i}$. A sufficient condition for the existence of common retract hosts is the existence of products or coproducts.

In this talk, based on joint work with Fernando Lucatelli Nunes and Matthijs Vákár [1], we present the three relevant completions, the equivalence theorem, and sufficient conditions where it holds. Finally, we discuss the originating motivation of the work in the context of categorical semantics of programming languages.

Keywords: category theory; free completions; Karoubi envelope
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References

[1] F. Lucatelli Nunes, D. Simm, and M. Vákár, *Simply Typed Reverse-Mode Automatic Differentiation with Variants: Denotational Correctness via Idempotent Completion*, 2026, arXiv:2607.14453. Available at: https://arxiv.org/abs/2607.14453